

F1 Dashboard

Software Engineering and Design Patterns

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10:25
5.08.20 17

Project Description

Problem: F1 telemetry is difficult to interpret

Goal: Interactive dashboard for driver comparison

F1 Telemetry

Year
2025

Race / Event
Monaco Grand Prix

Session
R

Load Session

Load Full Event

2025 Monaco Grand Prix – R

Loaded session from local database.

P1 — WINNER Lando Norris McLaren	P2 Charles Leclerc Ferrari	P3 Oscar Piastri McLaren	⚡ FASTEST LAP NOR 73.221 McLaren
Selected Laps: 78	Max Speed: 289.0 km/h	Samples: 22400	Best Lap: 73.221 s

End-users

- Formula 1 fans
- Sim racing enthusiasts
- Users interested in interactive telemetry analysis

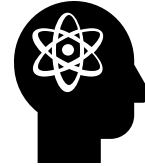


Roles



Spyridon
Product Owner

- Product vision & requirements
- Architecture decisions



Mariia
Scrum Master

- Agile workflow coordination
- Progress & milestone management



Kai
Developer

- Dashboard prototype & debugging
- Feature support & testing



Pia
Developer

- Dashboard implementation
- Database & telemetry integration

Features

Year

2020

Race / Event

British Grand Prix

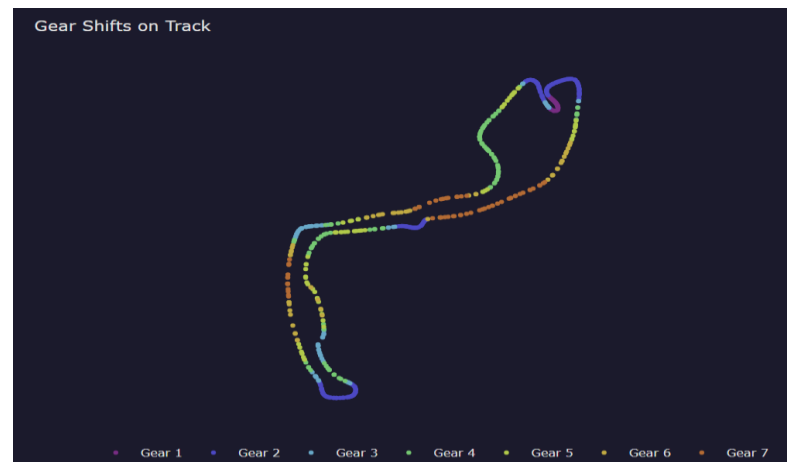
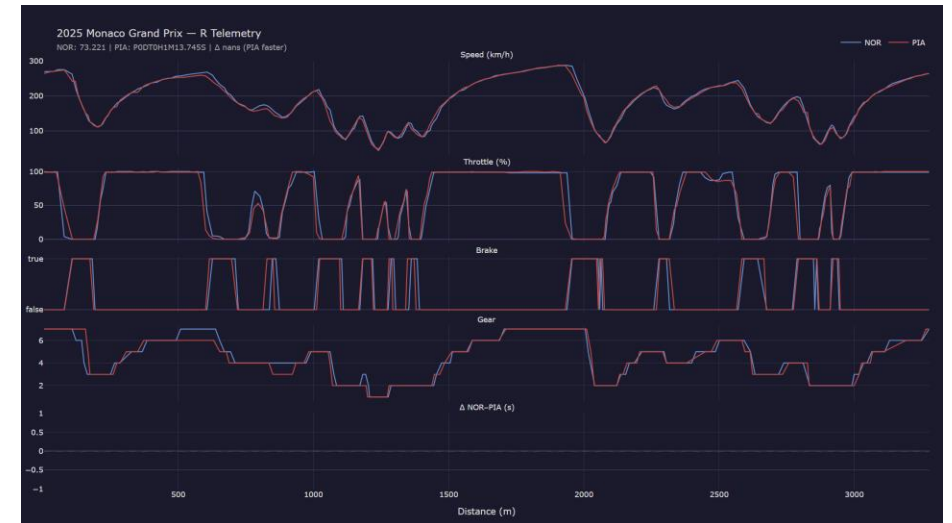
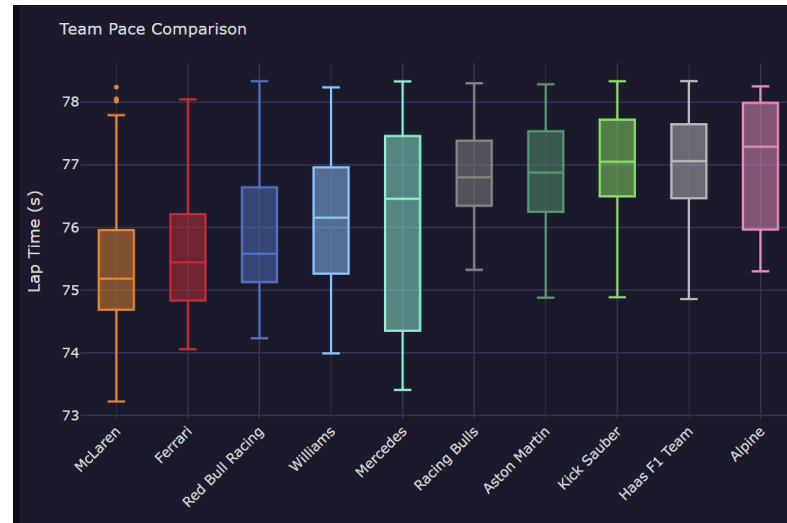
Session

R

Load Session

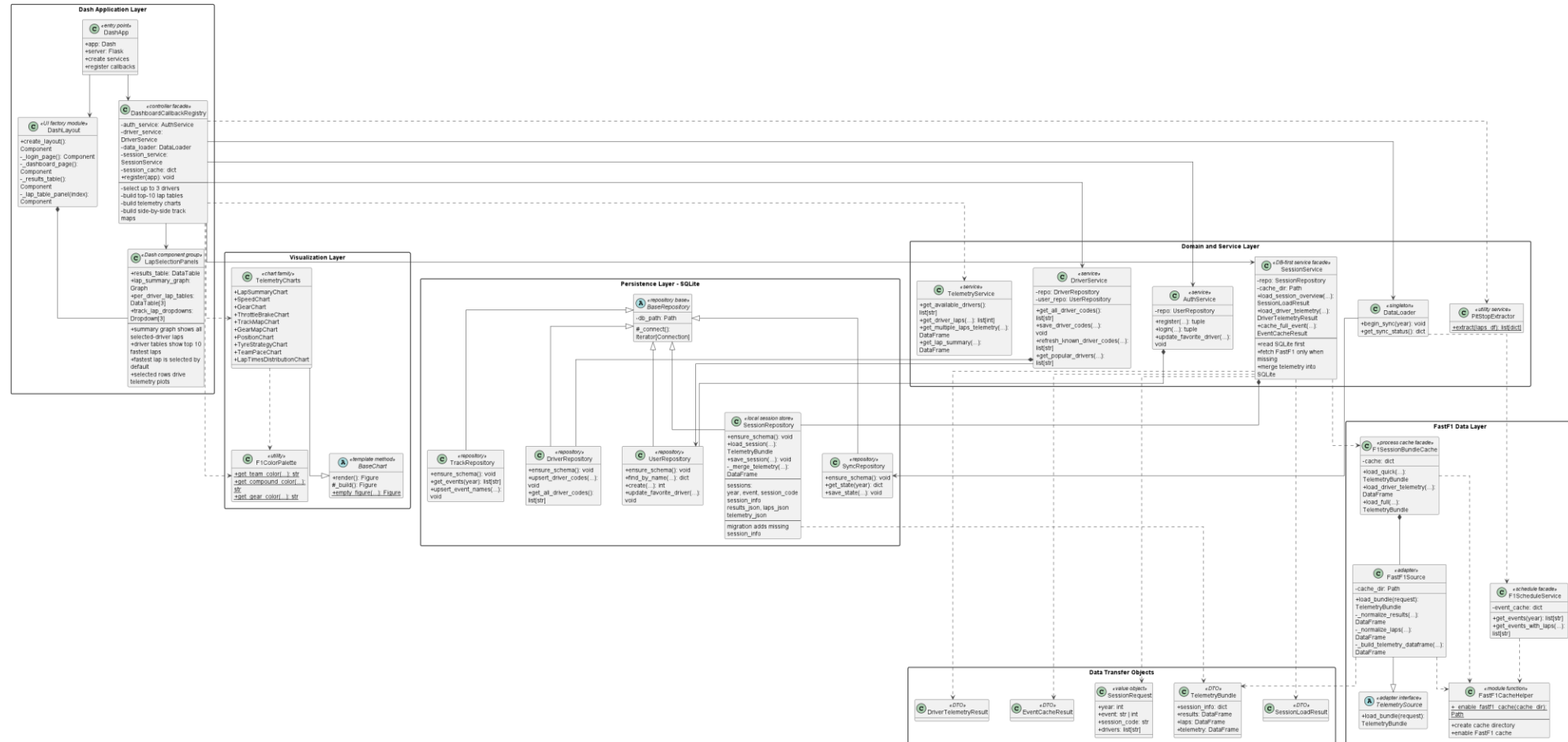
Load Full Event

Loaded: FP1, FP2, FP3, Q, R | Unavailable: SQ, S | Telemetry still loads per selected driver.



System Architecture – UML

F1 Telemetry Dashboard - Detailed UML Description



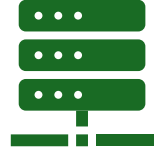
Arrow	Meaning
Solid arrow	Main runtime flow / direct collaboration
Dashed arrow	Secondary dependency, helper, or generated data
Open diamond	Composition / owned object
Half-filled triangle	Inheritance or interface implementation
Dash -> Domain	Callbacks use injected services for auth, drivers, sync, sessions, metrics, and pit-stop extraction
Domain -> SQLite	Services own repositories for user, driver, track, sync, and local session data
Domain -> FastF1	SessionService uses FastF1 only when SQLite is missing overview or driver telemetry
Domain -> Visualization	Callbacks build pretty figures through chart classes and F1ColorPalette
Finishing order selects drivers, lap-to-lap tables select telemetry laps, track drop-downs select map/laps	
Generated outputs	DTM, DTMD renders live in dashboard

System Architecture – Layers



Dash Application Layer:

- App setup
- Layout
- Input controls
- Callback wiring



Domain & Service Layer:

- Authentication
- Session loading
- Data loading
- Background caching



Data Persistence Layer:

- Local data storage
- Users, favorite drivers
- Telemetry data
- Metadata



FastF1 Data Layer:

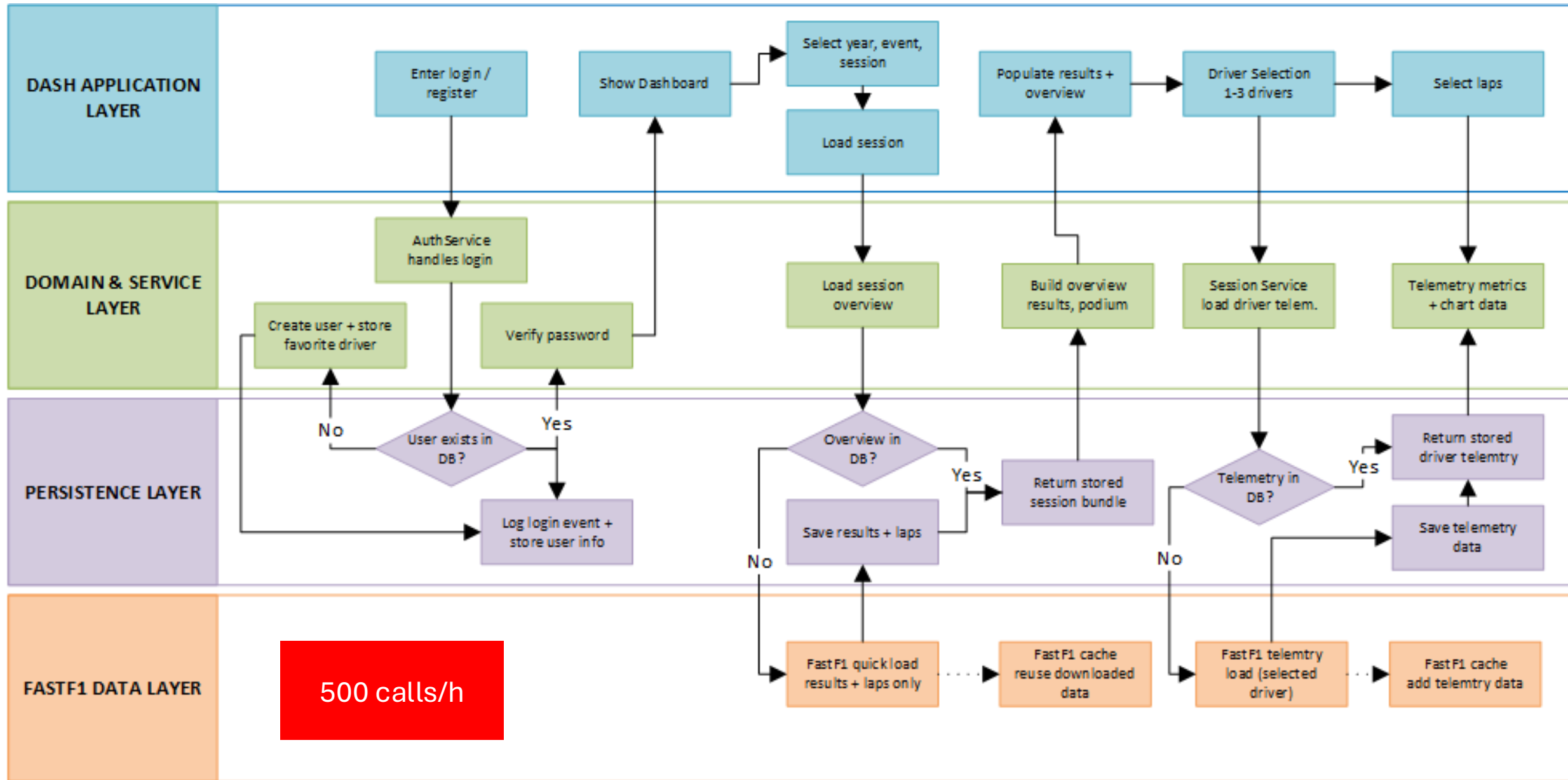
- Loads data from FastF1
- Enables caching
- Structures data for Service Layer



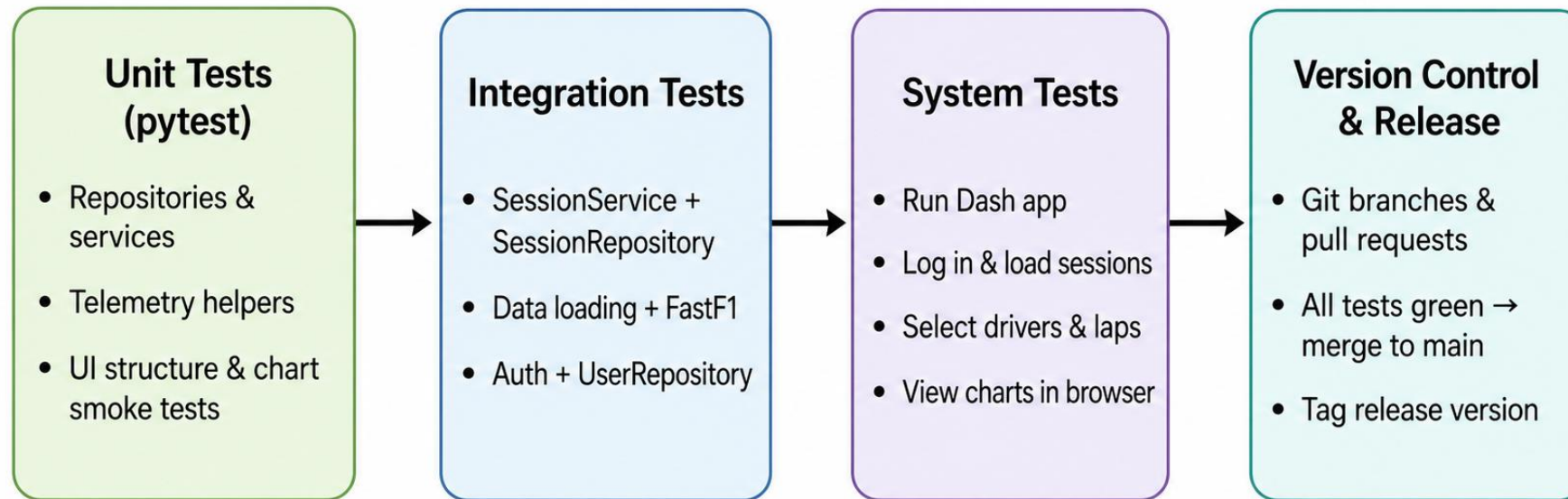
Visualization Layer:

- Builds Plotly figures
- Position charts
- Track maps
- Speed charts

System Architecture – Swimlane



Quality Assurance Process



Year

2025

Race / Event

Abu Dhabi Grand Prix

Session

R

Load Session

Load Full Event

Full event loads all available session overviews for this Grand Prix.

2025 Abu Dhabi Grand Prix – R

Loaded session from local database.

Race Overview

P1 - WINNER

Max Verstappen

Red Bull Racing

P2

Oscar Piastri

McLaren

P3

Lando Norris

McLaren

FASTEST LAP

LEC 86.725

Ferrari

Selected Laps: 2

Max Speed: 309.0 km/h

Samples: 637

Best Lap: 86.765 s

Live Demo

Race Timeline – Position Chart



Key Events

- HUL PIT Lap 7

ALB PIT Lap 8

HAM PIT Lap 8

GAS PIT Lap 13

BEA PIT Lap 14

HAD PIT Lap 14

RUS PIT Lap 14

BOR PIT Lap 15

COL PIT Lap 15

ALO PIT Lap 16

LEC PIT Lap 16

NOR PIT Lap 16

SAI PIT Lap 18

OCO PIT Lap 19

LAW PIT Lap 21

VER PIT Lap 23

HAM PIT Lap 31

ANT PIT Lap 32

TSU PIT Lap 32

ALB PIT Lap 33

Thank you

Do you have questions ?

